Coding Agent Evaluation

Project: PathSG / 2006_SCSL_Team2Project
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This report covers the reconstruction and review of that version. Current test results are recorded in ../testing/Black_Box_Testing.md.

1. Profile Controller Reconstruction

Inputs and Request

We chose UserProfileController, whose code is in client/src/stores/profile.ts and client/src/utils/profileValidation.ts. We gave Codex the existing tests, profile view, types, education definitions, postal utilities, and code that calls the controller. The first attempt did not use the requirements document or class diagram directly.

We asked Codex to rebuild both files after removing them, keep the existing interfaces and behavior, and run the focused tests, regression suite, type check, and build.

Procedure

1. We backed up both files and recorded their SHA-256 hashes. All 63 focused tests passed before the removal.
2. We removed the files. The tests could not load because the imports were missing.
3. Codex generated replacements using the tests and surrounding code. All 63 focused tests and 610 frontend tests passed.
4. We compared the replacements with the backup and found differences in number formats, one error message, and when messages were cleared.
5. We added ten tests for those differences. All ten failed against the first generated validator.
6. Codex corrected the format checks, error message, and watchers. The score-label lookup was also changed to a type-safe partial mapping. The final tests passed.

Results

Stage	Command from client	Result
Baseline	npm test -- profileValidation.spec.ts profile.spec.ts	63 passed
Files removed	Same focused command	2 suites failed to load; no tests executed
First reconstruction	Same focused command	63 passed
First regression	npm test	610 passed
Initial types	npm run type-check	Passed
Additional format checks	npm test -- profileValidation.spec.ts	10 failed; 53 passed
Corrected focused run	npm test -- profileValidation.spec.ts profile.spec.ts	73 passed
Corrected regression	npm test	620 passed
Final types	npm run type-check	Passed
Changed-file formatting	Prettier check	Passed
Production build	npm run build-only	Passed

We checked the final version against the original code as well as the tests.

2. Code Review

We also asked Codex to review parts of the frontend, backend, and CI workflow. The brief asked for problems tied to specific files, an explanation of their impact, and a prioritised list of refactoring suggestions. Codex read the frontend stores and services, backend models, routes and authentication code, and the CI configuration. The main findings for that version were:

Priority	Finding	Recommendation
High	Some frontend services still simulated responses	Connect them to backend endpoints and test identifiers and payloads
High	The admin interface relied on a client-side check	Check user roles at privileged server endpoints
Medium	Roadmap save used separate lookup/write sessions	Use one transaction and handle uniqueness conflicts
Medium	Client/server timeout values differed	Align timeout behavior and test failure handling
Medium	Pagination inputs lacked explicit nonnegative bounds	Define ranges and reject invalid queries
Low	The score-label lookup used a type assertion	Use Partial<Record<TierId, string>>; applied during reconstruction

We ran the frontend checks but did not run the backend or live services for this review. The larger changes remained recommendations.

3. Reflections

Reconstruction

The first version passed the existing tests, but the comparison with the backup found differences that those tests had missed. Adding the ten format and message tests made the problem clear. After the corrections, 73 focused tests and all 620 frontend tests passed. We found that checking the generated code against existing behavior was just as useful as running the original tests.

We expected the tests and surrounding code to be enough to recover the interface and main behavior. The first attempt met that aim, but matching the number formats and messages needed the extra tests and follow-up corrections.

Code Review

The review gave us specific issues to check, including service integration, access control, roadmap saves, and timeouts. We checked the suggestions against the code and CI workflow before deciding what to change. We applied the small typing fix; the larger changes needed more work and were left as recommendations.

The review met our aim of getting a practical list of improvements. Follow-up checking helped separate useful changes from suggestions already covered by the existing code or CI setup. The applied typing change was small; the other findings need their own implementation and integration tests.